WORKED SOLUTION

STEP II 2011, Question 5

Pure • Vectors and Reflections

STEP II 2011, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s2-q5

The key idea

To reflect a point in a line through the origin, split its position vector into a part along the line and a part perpendicular to it: the reflection keeps the first and reverses the second. The component of $\mathbf{b}$ along $\mathbf{a}$ is the familiar projection $\frac{\mathbf{a}\cdot\mathbf{b}}{\mathbf{a}\cdot\mathbf{a}}\,\mathbf{a}$, and everything else is bookkeeping with dot products. For the collinearity, remember that a point lies on line AB exactly when its position vector is $s\mathbf{a}+t\mathbf{b}$ with $s+t=1$.

Solution

Throughout, $\mathbf{a}$ and $\mathbf{b}$ are non-zero and not parallel, because O, A, B are not collinear. In particular $\mathbf{a}\cdot\mathbf{a}\neq 0$ and $\mathbf{b}\cdot\mathbf{b}\neq 0$, so we may divide by them. Write $\theta=\angle AOB$.

The reflection C

Let P be the foot of the perpendicular from B to the line OA. Its position vector is a multiple $k\mathbf{a}$, and $\mathbf{b}-k\mathbf{a}$ is perpendicular to $\mathbf{a}$:

$$(\mathbf{b}-k\mathbf{a})\cdot\mathbf{a}=0 \quad\Longrightarrow\quad k=\frac{\mathbf{a}\cdot\mathbf{b}}{\mathbf{a}\cdot\mathbf{a}}.$$

Reflecting in the line OA fixes P and sends B to the point C on the other side with P as the midpoint of BC. So $\mathbf{c}=2\overrightarrow{OP}-\mathbf{b}$, that is,

$$\mathbf{c}=\frac{2\,\mathbf{a}\cdot\mathbf{b}}{\mathbf{a}\cdot\mathbf{a}}\,\mathbf{a}-\mathbf{b}=\lambda\mathbf{a}-\mathbf{b}, \qquad \lambda=\frac{2\,\mathbf{a}\cdot\mathbf{b}}{\mathbf{a}\cdot\mathbf{a}},$$

as required.

The reflection D

The same argument with the line OB in place of OA gives

$$\mathbf{d}=\frac{2\,\mathbf{b}\cdot\mathbf{c}}{\mathbf{b}\cdot\mathbf{b}}\,\mathbf{b}-\mathbf{c}.$$

Now $\mathbf{b}\cdot\mathbf{c} = \lambda\,\mathbf{a}\cdot\mathbf{b} - \mathbf{b}\cdot\mathbf{b}$, so

$$\begin{aligned}\mathbf{d} &= \left(\frac{2\lambda\,\mathbf{a}\cdot\mathbf{b}}{\mathbf{b}\cdot\mathbf{b}} - 2\right)\mathbf{b} - (\lambda\mathbf{a} - \mathbf{b}) \\ &= \left(\frac{2\lambda\,\mathbf{a}\cdot\mathbf{b}}{\mathbf{b}\cdot\mathbf{b}} - 1\right)\mathbf{b} - \lambda\mathbf{a}.\end{aligned}$$

This is $\mathbf{d} = \mu\mathbf{b} - \lambda\mathbf{a}$ with

$$\mu = \frac{2\lambda\,\mathbf{a}\cdot\mathbf{b}}{\mathbf{b}\cdot\mathbf{b}} - 1 = \frac{4(\mathbf{a}\cdot\mathbf{b})^2}{(\mathbf{a}\cdot\mathbf{a})(\mathbf{b}\cdot\mathbf{b})} - 1 = 4\cos^2\theta - 1,$$

using $\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$ in the last step. The form $4\cos^2\theta - 1$ is the one we shall need below.

A, B, D collinear

The point D lies on the line AB if and only if $\mathbf{d} = \mathbf{a} + t(\mathbf{b} - \mathbf{a}) = (1-t)\mathbf{a} + t\mathbf{b}$ for some scalar t. We also have $\mathbf{d} = -\lambda\mathbf{a} + \mu\mathbf{b}$.

Ask yourself Why can we compare the coefficients of $\mathbf{a}$ and $\mathbf{b}$?

Because $\mathbf{a}$ and $\mathbf{b}$ are not parallel. If $s\mathbf{a} + t\mathbf{b} = s'\mathbf{a} + t'\mathbf{b}$ then $(s - s')\mathbf{a} = (t' - t)\mathbf{b}$, and unless both brackets are zero this would make $\mathbf{a}$ and $\mathbf{b}$ parallel. This is where the non-collinearity of O, A, B is used.

So $-\lambda = 1 - t$ and $\mu = t$. Eliminating t, A, B and D are collinear exactly when

$$\mu = \lambda + 1.$$

The case $\lambda = -\frac{1}{2}$

Collinearity now gives $\mu = \frac{1}{2}$, so $4\cos^2\theta - 1 = \frac{1}{2}$ and $\cos^2\theta = \frac{3}{8}$. Since $\lambda < 0$ we have $\mathbf{a}\cdot\mathbf{b} < 0$, so $\cos\theta < 0$ and

$$\cos\angle AOB = -\sqrt{\tfrac{3}{8}} = -\frac{\sqrt{6}}{4}.$$

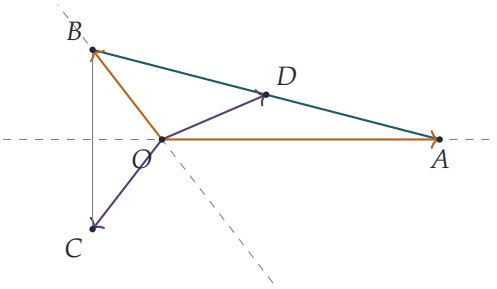
Then

$$\mathbf{d} = \tfrac{1}{2}\mathbf{b} + \tfrac{1}{2}\mathbf{a} = \tfrac{1}{2}(\mathbf{a} + \mathbf{b}),$$

so D **is the midpoint of** AB. We can say more about the shape. From $\lambda = \dfrac{2|\mathbf{b}|\cos\theta}{|\mathbf{a}|} = -\frac{1}{2}$,

$$\frac{|\mathbf{a}|}{|\mathbf{b}|} = -4\cos\theta = \sqrt{6},$$

so $OA = \sqrt{6}\,OB$, and the angle AOB is obtuse (about 127.8°). Also, reflections preserve distance from O, so $OD = OC = OB$: the midpoint of AB is exactly as far from O as B is.



Checks

- Reflecting twice, first in OA and then in OB, is a rotation about O through 2θ. So D should be B rotated through 2θ, with $\cos\angle BOD = \cos 2\theta = 2\cos^2\theta - 1$. In the final case this is $2\cdot\frac{3}{8} - 1 = -\frac{1}{4}$. Directly, $\mathbf{b}\cdot\frac{1}{2}(\mathbf{a}+\mathbf{b}) = \frac{1}{2}(-\frac{3}{2}+1)|\mathbf{b}|^2 = -\frac{1}{4}|\mathbf{b}|^2$, using $\mathbf{a}\cdot\mathbf{b} = -\frac{1}{4}\,\mathbf{a}\cdot\mathbf{a} = -\frac{3}{2}\,\mathbf{b}\cdot\mathbf{b}$, which agrees.
- If $\theta = 90°$ then $\lambda = 0$, C is $-\mathbf{b}$ and D is $-\mathbf{b}$ again. The formula gives $\mu = 4\cdot 0 - 1 = -1$, as it should.
- Numerically, with $\mathbf{a} = (\sqrt{6}, 0)$ and $\mathbf{b} = (-0.6124, 0.7906)$, reflecting twice gives $\mathbf{d} = (0.9186, 0.3953)$, which is the midpoint of AB, and $|\mathbf{d}| = 1 = |\mathbf{b}|$.

Common slips

- Writing the reflection as $\mathbf{b} - 2\times(\text{projection})$. That reflects in the line *perpendicular* to OA. Check by testing $\mathbf{b} = \mathbf{a}$, which should be fixed.
- Taking $\cos\theta = +\sqrt{6}/4$. The sign of λ fixes the sign of $\mathbf{a}\cdot\mathbf{b}$.
- Comparing coefficients without saying that $\mathbf{a}$ and $\mathbf{b}$ are independent.

Beyond STEP

The map $\mathbf{x} \mapsto \frac{2\,\mathbf{a}\cdot\mathbf{x}}{\mathbf{a}\cdot\mathbf{a}}\,\mathbf{a} - \mathbf{x}$ is a linear map, and its matrix is a standard first-year example of an orthogonal matrix with determinant -1. Composing two reflections gives a rotation through twice the angle between the mirrors. This is the start of the study of isometries and of groups generated by reflections, such as the symmetry groups of regular polygons. Strang's *Introduction to Linear Algebra* covers projections and reflection matrices well.

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